# Multi-Facility Patient Flow Optimization: A Simulation Approach

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**Abstract**

Emergency department (ED) boarding, where admitted patients remain in the ED due to unavailable inpatient beds, contributes to prolonged wait times, degraded care quality, and increased patient mortality. In multi-hospital health systems, understanding patient flow across facilities is essential for identifying bottlenecks and evaluating system-wide capacity. This project develops a multi-facility discrete event simulation (DES) of patient flow across UC San Diego Health's three campuses (La Jolla, Hillcrest, and East Campus) using institutional Admit-Discharge-Transfer (ADT) data comprising 2.1 million events across 985,000 patient encounters over a recent one-year observation period.

We define a 12-state model consisting of four care tiers (ED, Floor, ICU, and Boarding) across three campuses and construct a trajectory library of 150,337 real patient paths, where each simulated patient follows an actual historical trajectory with empirical durations. Baseline results identify La Jolla and East Campus as the most chronically constrained facilities, with boarding and ICU resources exceeding 80% utilization for the majority of the simulation period. We then integrate a quantile regression LOS predictor as a transfer trigger and evaluate policies across three thresholds (3, 8, and 12 hours). LOS-triggered transfers reliably eliminate boarding congestion at the source campus, but system-level benefit depends on the choice of source: transfers from La Jolla produce unambiguous improvement, while transfers from smaller campuses introduce load-shifting effects that can amplify bottlenecks at receiving facilities.

Website: https://nadineorriss.github.io/simhospital-website/
Code: https://github.com/kayannet/SimHospital

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# 1 Introduction

Boarding in the emergency department (ED), when admitted patients remain in the ED due to unavailable inpatient beds, contributes to prolonged wait times, degraded care quality, and increased patient mortality. Effective capacity planning must address two fundamental questions: First, how many beds does a hospital actually need to handle typical patient volume across its care tiers? Second, can predictive models identify patients requiring extended hospital stays, enabling early transfer to secondary facilities to reduce ED boarding?

While the causes of ED crowding are multifaceted (patient influx, hospital bed availability, discharge bottlenecks), traditional approaches rely on historical peak occupancy ("we maxed out at 50 beds during flu season, so we need at least 50"). This reactive heuristic fails to account for demand variability, differences between average and peak utilization, downstream bottlenecks in the hospital admission process, and tradeoffs between capital investment (adding beds) versus operational efficiency (improving patient flow through transfers). In multi-hospital health systems, these challenges are compounded by the need to coordinate capacity across campuses: an ICU at capacity at one facility may force patients to board in the ED, while a neighboring campus with available beds goes underutilized. Understanding these cross-facility dynamics is critical for planning resource reallocation during operational disruptions such as nursing strikes, seasonal surges, or facility closures.

Discrete-event simulation (DES) provides a systematic method for analyzing these dynamic, interconnected factors. By modeling patient flow as a sequence of timestamped events (arrival, triage, treatment, transfer, disposition), DES enables researchers to estimate capacity requirements, test operational interventions, and predict performance under alternative scenarios. Prior studies have demonstrated DES applications in ED workforce planning Day et al. (2012), process optimization Ahmed and Alkhamis (2009), and bottleneck identification Kuo et al. (2014).

However, most hospital simulations suffer from two critical limitations: (1) Limited reproducibility, as studies often use site-specific, proprietary data unavailable to other researchers, making validation and extension difficult. (2) Single-facility scope, as the majority of published DES models simulate a single ED or hospital unit in isolation, ignoring the inter-facility transfers and shared capacity constraints that characterize real health systems. This project addresses the second limitation by modeling patient flow across three campuses simultaneously. However, the use of institutional data necessarily limits external reproducibility, a tradeoff we accept in exchange for the ability to calibrate and validate the simulation against real operational patterns.

This project develops a multi-facility discrete event simulation of patient flow across UC San Diego Health's three hospital campuses (La Jolla, Hillcrest, and East Campus). Patients entering the system may be seen in the ED, admitted to inpatient floor units, escalated to the ICU, transferred between campuses, or discharged at various points along their care pathway. We model this system as a 12-state simulation consisting of four care tiers (ED, Floor, ICU, and Boarding) across three campuses, plus discharge. Rather than generating synthetic patient paths from theoretical distributions, the simulation is driven by a trajectory library of over 150,000 real patient journeys extracted from institutional ADT data,

preserving the complexity of actual clinical pathways. To evaluate dynamic inter-campus transfer policies, we integrate a length-of-stay quantile regression predictor that identifies patients likely to require extended stays at the time of ED admission, triggering rerouting to alternative campuses under a range of LOS thresholds.

## 1.1 Literature Review

Discrete-event simulation (DES) has been widely applied to model emergency department (ED) operations and evaluate improvement strategies. Day et al. (2012) used DES to evaluate the impact of adding a healthcare provider to triage at a Veterans Affairs ED, demonstrating a 15% reduction in mean length of stay (LOS) alongside a 28% decrease in patients with LOS exceeding 6 hours. Similarly, Ahmed and Alkhamis (2009) developed a simulation–optimization model to optimize staffing levels in a Kuwaiti ED, showing that strategic resource allocation could achieve a 40% reduction in patient waiting time and a 28% increase in throughput without increasing costs.

Prior work has increasingly focused on using real-world data to parameterize simulation models. Kuo et al. (2014) used electronic health record data to develop service-time distributions through indirect imputation for various ED processes, improving model fidelity compared to traditional assumption-based approaches. Building on these methods, Boyle et al. (2021) presented a data-driven framework for developing generalisable DES models of hospital emergency departments, emphasizing the use of hospital information system data to ensure model credibility and support reuse across multiple sites. Paul et al. (2010) conducted a systematic review emphasizing the importance of validating DES outputs against actual hospital metrics, noting that poorly calibrated models can lead to misleading policy recommendations.

Clearly, most ED simulations face a tradeoff between reproducibility and validation. Studies using proprietary data can validate against operations but cannot be reproduced; studies using public data enable reproducibility but cannot validate without institutional collaboration. Additionally, the majority of published work focuses on single-facility modeling, leaving multi-campus patient flow largely unaddressed.

## 1.2 Data Description

This project uses de-identified Admit-Discharge-Transfer (ADT) event data from UC San Diego Health's Joint Commission on Hospital Improvement (JCHI) data extract. The dataset spans a one-year observation period and contains approximately 2.1 million ADT event records across 985,000 patient encounters documented at three hospital campuses.

Each encounter is tracked through timestamped events including admissions, transfers between units, census snapshots, patient updates, and discharges. The raw data structure records one row per event, with each event associated with a specific patient encounter, department, care tier, and timestamp. The dataset contains 37 columns organized into several categories: patient identifiers (medical record number and encounter ID), event

information (event type, timestamp, sequence number), transfer tracking (linked transfer event IDs and transfer type), location information (department name, department group, room, and bed identifiers), patient classification (base patient class and detailed patient class), clinical service (hospital service line), and bed request tracking fields.

Six event types are recorded: Admission, Transfer In, Transfer Out, Discharge, Census, and Patient Update. Transfer events are linked through shared event IDs, allowing reconstruction of patient movement between departments. The department group field categorizes units into nine clinical groupings: MedSurg, OB, PCU, ICU, IMU, MedSTele, Behavioral Health, Nursery, and NICU. Notably, department group assignments are missing for emergency department events in the source data, as this field was never populated for EDs. These events are instead identified through patient class labels and department name prefixes, a distinction that informed our data cleaning pipeline described in Section 2.1.

The raw data includes a mix of inpatient, emergency, and outpatient encounters. Outpatient visits such as radiology, infusion therapy, and clinic appointments account for the majority of encounters but do not involve physical bed occupancy or meaningful patient flow. Filtering these encounters to isolate bed-occupying patients with modelable transitions is a central component of our preprocessing methodology.

# 2 Methods

## 2.1 Data Preprocessing

The raw ADT dataset required substantial cleaning to isolate encounters relevant to patient flow modeling. We developed a four-step preprocessing pipeline to transform 2.1 million raw event records into simulation-ready data.

### 2.1.1 Step 1: Filter to Bed-Occupying Patients

The raw data contains a mix of inpatient, emergency, and outpatient encounters. The majority of encounters are outpatient visits (radiology, infusion therapy, procedures, clinic visits) that do not occupy physical hospital beds and produce no meaningful patient flow to model. We identified 11 patient classes representing patients who occupy beds and can transfer between units: Inpatient Admission, Emergency Department Encounter, Trauma Inpatient Admission, Observation, Trauma Observation Admission, Hospital Inpatient Surgery, Inpatient Psychiatry, Newborn, Surgical OB Patient, Outpatient in a Bed (patients physically occupying ICU/MedSurg/PCU beds despite outpatient classification), and Outpatient Labor and Delivery Encounter (patients in OB units). The remaining 22 patient classes, all representing same-day outpatient services with no bed occupancy, were removed. This reduced the dataset from 2,187,128 rows (984,881 encounters) to 1,153,186 rows (153,878 encounters), removing 47.3% of rows while retaining all encounters with physical patient flow.

### 2.1.2 Step 2: Drop Single-Event Encounters

After filtering, 1,720 encounters (1.1%) contained only a single event, producing zero-length stays and no usable transitions. Investigation confirmed these were not artifacts of our filtering but incomplete records in the source data (orphan discharges, registration-only entries) with a mean of only 1.08 events per encounter in the original unfiltered data. These were dropped, yielding 152,158 encounters and 1,151,386 events.

### 2.1.3 Step 3: Tier Mapping

Our simulation models patient flow between four care tiers (ED, Floor, ICU, and Boarding) across three campuses (La Jolla, Hillcrest, East Campus). The raw data uses approximately ten granular department groups (MedSurg, OB, PCU, ICU, IMU, MedSTele, Behavioral Health, Nursery, NICU) that needed to be collapsed into these four tiers. We applied the following mapping:

- **ED**: Events where patient class is "Emergency Department - Encounter" or where the department name contains an emergency department prefix (LJ/HC/EC EMERGENCY DEPT). ED events had missing department group values in the source data, as this field was never populated for emergency departments.
- **Boarded**: Events where patients remained in the emergency department after admission while awaiting an inpatient bed. Identified when the patient class is not "Emergency Department - Encounter" but the department group is "Emergency."
- **ICU**: Events where department group is "ICU."
- **Floor**: All remaining events with a non-null department group (MedSurg, OB, PCU, IMU, MedSTele, Behavioral Health, Nursery, NICU).

Events with no assignable tier (perioperative areas, holding rooms, recovery rooms) were assigned a null tier. These events represent locations outside the ED/Floor/ICU care tier system and are skipped during transition extraction, with transitions connecting around them. For example, a patient path of ED → periop → Floor becomes ED → Floor, preserving the meaningful transition without introducing an intermediate state that does not correspond to a modeled resource.

Each event was also assigned a campus based on the department name, and the tier and campus were combined into a state label (e.g., "ED_LA_JOLLA", "FLOOR_HILLCREST", "ICU_EAST_CAMPUS").

### 2.1.4 Step 4: Timestamp-Based Ordering

The sequence number field in the source data (`seq_num_in_enc`) is scoped to the Inpatient Admission patient class rather than the full encounter. When a patient starts in the ED and converts to inpatient status, the initial ED admission event receives no sequence number. Since all such cases follow the same pattern and timestamps are in correct chronological order, we use the `effective_time` field to order all events within each encounter. No

resequencing was required.

## 2.2 State Space Definition

The simulation defines thirteen possible states: twelve active states representing each combination of care tier and campus, plus a discharge state. The twelve active states are:

| | La Jolla | Hillcrest | East Campus |
|---|---|---|---|
| ED | ED_LJ | ED_HC | ED_EC |
| Floor | Floor_LJ | Floor_HC | Floor_EC |
| ICU | ICU_LJ | ICU_HC | ICU_EC |
| Boarded | Boarded_LJ | Boarded_HC | Boarded_EC |

Each active state corresponds to a pool of beds modeled as a resource in the simulation. A patient occupies exactly one state at a time and transitions between states according to their trajectory. Boarded patients are those who have been admitted as inpatients but physically remain in the ED awaiting an available inpatient bed, representing a clinically distinct state from active ED patients. Modeling boarding as an explicit state rather than inflating ED occupancy allows the simulation to track boarding bed demand, queue pressure, and wait times independently from active ED patients, reflecting the operational reality that boarded and active ED patients compete for distinct physical and administrative resources. All patient journeys end in the discharge state.

## 2.3 Trajectory Library Construction

Rather than using transition probabilities to stochastically generate patient paths, we constructed a trajectory library containing the actual sequence of states and durations observed in the historical data. This approach preserves the full complexity of real clinical pathways, including rare but important patterns such as cross-campus transfers or repeated ICU escalations that might be underrepresented in a probability-based model.

For each encounter, we extracted the sequence of states visited (after dropping null-tier events and removing consecutive duplicate states) along with the duration spent in each state. Duration was calculated as the difference between the first and last event timestamps within each contiguous block of the same state. Each encounter was then represented as an ordered list of (state, duration) tuples. For example, a patient who spent 4.5 hours in the La Jolla ED followed by 48.2 hours on the La Jolla floor would be represented as:

```
ED_LA_JOLLA (4.5h) → FLOOR_LA_JOLLA (48.2h) → Discharge
```

The final trajectory library contains 150,337 patient trajectories, organized by starting campus. Of the 152,158 encounters remaining after preprocessing, 1,821 produced no valid state transitions after null-tier removal and were excluded. During simulation, each arriving patient is assigned a trajectory by randomly sampling from the library of real encounters that began at the same campus.

## 2.4 Arrival Rate Modeling

Patient arrival patterns differ between weekdays and weekends, with weekends showing 10–25% fewer arrivals across all campuses, consistent with reduced elective admissions and different ED volumes. We calculated mean daily arrivals by campus and day type using the first event per encounter as the arrival timestamp.

| Campus | Weekday (mean ± std) | Weekend (mean ± std) |
|---|---|---|
| East Campus | 92.4 ± 10.5 | 83.3 ± 10.5 |
| Hillcrest | 144.7 ± 19.9 | 125.1 ± 10.3 |
| La Jolla | 198.2 ± 26.2 | 148.9 ± 14.5 |

In the simulation, arrivals at each campus are generated as a Poisson process. The mean inter-arrival time is computed as 24 hours divided by the mean daily arrival count for the appropriate day type. The simulation checks the current simulation clock to determine whether the current day is a weekday or weekend and selects the corresponding arrival rate, producing time-varying arrival patterns that reflect real operational rhythms.

## 2.5 Simulation Framework

The discrete event simulation was implemented in Python using the SimPy library for process-based simulation and the vidigi library for event logging and resource tracking. The simulation was run for 4,328 hours (approximately six months) per run.

### 2.5.1 Architecture

The simulation is organized around four core classes:

**Hospital**: Each campus is represented as a Hospital object that owns and manages all bed resources for that campus. The Hospital class initializes VidigiStore resource pools for each of the four care tiers (ED, Floor, ICU, Boarded), stores per-state fitted LOS distribution parameters, and tracks median LOS per resource derived from the trajectory library. This campus-level encapsulation allows the simulation to manage resources, arrival rates, and LOS parameters independently per facility.

**Patient**: Each simulated patient is initialized with a unique identifier, an encounter ID linking to the sampled trajectory, and the full path (list of state-duration tuples) from the trajectory library. The patient object tracks the current state, wait times and service times at each state, total time in the system, transfer history, and clinical flags.

**Model**: The Model class creates a SimPy environment and instantiates one Hospital object per campus. The Model contains three key methods:

- `generator_patient_arrivals(campus)`: Runs continuously for each campus. At each step, it randomly samples an encounter from the trajectory library for that campus, creates a Patient with the sampled path, starts the patient journey process,

and waits for an exponentially distributed inter-arrival time based on the current weekday or weekend arrival rate.

- `patient_journey(patient)`: Iterates through the patient's trajectory. At each state, the patient requests a bed from the corresponding resource pool within the appropriate Hospital object. If beds are available, the patient immediately begins occupying the resource; if not, the patient waits in a queue. Once a bed is obtained, the patient occupies it for the duration specified in the trajectory, then releases it and moves to the next state. When the transfer policy is active, this method additionally evaluates whether a transfer should be triggered at the moment of ED admission, as described in Section 2.8.
- `interval_audit_utilization`: A background process that records the number of beds in use and queue lengths for all resources across all campuses at regular intervals (every simulated hour), producing time-series utilization data.

**Trial**: The Trial class manages multiple simulation runs, instantiating a new Model for each run and collecting results including mean queue times, mean service times per state, and complete event logs across all runs.

### 2.5.2 Capacity Calibration

Bed capacities for each of the twelve active states were derived empirically from an unconstrained simulation run in which all capacities were set to 9,999, effectively removing resource constraints and allowing patient demand to fully saturate the system without artificial queueing. After a warm-up period of 50 days, we computed the mean steady-state occupancy for each state, representing the average number of beds simultaneously occupied under real arrival rates and trajectory durations. These mean post-warmup occupancy values, rounded to the nearest integer, were used as the capacity parameters for all constrained simulation runs. This approach calibrates capacity to typical observed demand rather than licensed bed counts, which were unavailable at the tier level. Table 1 reports the resulting capacity parameters.

Table 1: Calibrated bed capacity parameters derived from steady-state occupancy in the unconstrained simulation run.

| Campus | ED | Floor | ICU | Boarded |
|---|---|---|---|---|
| La Jolla | 67 | 348 | 57 | 49 |
| Hillcrest | 31 | 218 | 30 | 10 |
| East Campus | 12 | 82 | 8 | 4 |

### 2.5.3 Fitted LOS Distributions

To support stochastic path generation during transfer rerouting, we fit parametric length-of-stay (LOS) distributions to empirical duration data extracted from the trajectory library. For each state, four candidate distributions were evaluated—lognormal, gamma, exponential,

and Weibull. The best-fitting distribution was selected by minimizing the sum of squared errors between the empirical histogram and the fitted probability density function.

Across all campuses, floor and ICU states were best fit by lognormal distributions, consistent with the heavy right-skew commonly observed in inpatient LOS data. Boarding states were best fit by exponential distributions, reflecting the relatively memoryless nature of boarding duration once patients have been admitted but are awaiting bed placement.

These fitted distributions are used exclusively during transfer rerouting to generate durations for newly simulated path segments, as described in Section 2.8.

## 2.6 Transition Probability Matrix

To support stochastic path generation during inter-campus transfer rerouting, we constructed an empirical state transition probability matrix from the cleaned ADT data. For each patient encounter, we extracted the sequence of states visited, removed consecutive duplicate states, and appended discharge as the terminal state. All consecutive state pairs (`from_state`, `to_state`) across all encounters were collected and tallied into a 12×13 count matrix covering all twelve active states plus discharge. Each row was then normalized by its row sum to produce transition probabilities, ensuring each row sums to one. The resulting matrix captures the empirical likelihood of a patient moving from any given state to any other state based on real observed patient flow, including cross-campus transfers. This matrix is used exclusively during transfer rerouting to stochastically sample the next destination state for patients whose predicted LOS exceeds the transfer threshold, as described in Section 2.8.

## 2.7 LOS Quantile Regression Predictor

To support early identification of patients likely to require extended hospital stays, we trained a set of quantile regression models to predict total length of stay (LOS) at the moment of inpatient admission. Quantile regression was chosen over mean prediction due to the pronounced right skew of hospital LOS distributions, where mean estimates are heavily influenced by outliers. By modeling the 25th, 50th, and 75th percentiles of LOS, the predictor provides a range of estimates reflecting uncertainty in predicted stay duration.

The response variable is total LOS in hours, log-transformed prior to model fitting to stabilize variance and improve model fit on the skewed distribution. Predictions are back-transformed via the exponential function to return estimates in hours. Three features are used:

| Feature | Description |
|---|---|
| `time_in_ed` | Total hours spent in the ED prior to admission |
| `num_prev_90_days` | Number of prior admissions in the preceding 90 days |
| `admit_hospital` | Admitting campus (La Jolla, Hillcrest, or East Campus) |

The admitting campus is included as a categorical feature via one-hot encoding, reflecting

observed differences in case mix and LOS distributions across UCSD Health's three facilities. Separate models are trained for each quantile ($q \in \{0.25, 0.50, 0.75\}$) and serialized for use within the simulation. Predictor performance is evaluated in Section 3.2.

## 2.8 Transfer Policy Simulation

To evaluate the effect of early inter-campus transfers on system-wide capacity utilization and patient wait times, we implemented a LOS-based transfer trigger within the simulation. Three separate experiments were conducted, one for each source campus (La Jolla, Hillcrest, and East Campus), each evaluated independently against a no-transfer baseline.

### 2.8.1 Transfer Trigger Logic

The transfer decision is evaluated at the moment a patient transitions from the ED to an inpatient state, representing the point of admission. At this moment, the simulation computes three features for the patient: total time spent in the ED (`time_in_ed`), number of prior admissions in the preceding 90 days (`num_prev_90_days`), and the admitting campus (`admit_hospital`). These features are passed to the 75th percentile quantile regression model to generate a predicted LOS. If the predicted LOS exceeds the threshold defined for the current simulation run, a transfer is triggered.

### 2.8.2 Transfer Rerouting

When a transfer is triggered, the patient's remaining trajectory is discarded and replaced with a stochastically generated path. The new path is constructed by iteratively sampling the next state from the empirical transition probability matrix (Section 2.6) until discharge is reached or the patient's remaining predicted LOS is exhausted. Durations for each new state are sampled from the fitted per-state LOS distributions (Section 2.5.3), capped at the patient's remaining LOS budget to prevent artificially extended stays.

### 2.8.3 Threshold Sweep and Experimental Design

Each source campus experiment was run across three LOS thresholds: 3, 8, and 12 hours. A threshold of 3 hours represents an aggressive transfer policy triggering early for most admitted patients, while 12 hours represents a more conservative policy targeting only patients with clearly extended predicted stays. Each experimental run was conducted over 4,328 simulated hours (approximately six months) with a fixed random seed of 42 for reproducibility. Results across all threshold conditions and source campuses are compared against the no-transfer baseline to quantify the effect of transfer policy aggressiveness on system-wide utilization, boarding pressure, and patient wait times.

# 3 Results

## 3.1 Simulation Validation

To assess whether the simulation produces operationally plausible resource utilization, we compare the calibrated capacity parameters derived from the unconstrained simulation run against publicly reported bed counts for each UCSD Health campus. UC San Diego Health reports total bed counts of 418 beds for La Jolla (Jacobs Medical Center and Sulpizio Cardiovascular Center combined), 381 beds for Hillcrest Medical Center, and 302 beds for East Campus Medical Center, for a system total of 1,101 beds.

Table 2 compares posted bed counts against the sum of calibrated capacity parameters across all four care tiers (ED, Floor, ICU, and Boarded) for each campus.

Table 2: Comparison of posted bed capacity against the sum of calibrated simulation capacity parameters across all care tiers.

| Campus | Posted Beds | ED | Floor | ICU | Boarded | Calibrated Total | Ratio |
|---|---|---|---|---|---|---|---|
| La Jolla | 418 | 67 | 348 | 57 | 49 | 521 | 1.25 |
| Hillcrest | 381 | 31 | 218 | 30 | 10 | 289 | 0.76 |
| East Campus | 302 | 12 | 82 | 8 | 4 | 106 | 0.35 |

Hillcrest and East Campus produce calibrated totals below posted capacity, which is expected since posted beds represent total licensed capacity rather than typical simultaneous occupancy—not all licensed beds are occupied at any given time. La Jolla's calibrated total exceeds posted capacity by 25%, attributable to the boarding effect: boarded patients physically occupy ED space but are counted as inpatients administratively, inflating the combined tier total beyond the licensed inpatient bed count. This is consistent with the known boarding dynamics at La Jolla, which is the highest-volume campus in the system. Overall, the calibrated parameters are consistent with expected operational patterns, providing confidence that the simulation reflects realistic utilization dynamics.

## 3.2 Predictor Performance

Figure 1 shows residual plots for each campus and quantile, where residuals are computed as actual minus predicted log LOS on the held-out test set. Across all three campuses and quantiles, residuals are broadly centered around zero, indicating no systematic directional bias in predictions.

The directional skew is consistent with the theoretical properties of quantile regression: at $q = 0.25$, residuals skew positive (most actual values exceed the 25th percentile prediction), while at $q = 0.75$, residuals skew negative (most actual values fall below the 75th percentile prediction), confirming that the models are correctly estimating the intended quantiles of the distribution.

A notable feature across all panels is the vertical banding pattern in the predicted values, reflecting the limited number of distinct predictions produced by a model with only three input features, one of which is categorical. While this limits predictive resolution, it is an expected consequence of using only information available at the moment of ED admission. The wide residual spread across all quantiles reflects the inherent difficulty of predicting LOS early in a patient's stay, consistent with prior literature on admission-time LOS prediction. East Campus shows a tighter residual spread than La Jolla and Hillcrest, consistent with its smaller and less variable patient population.

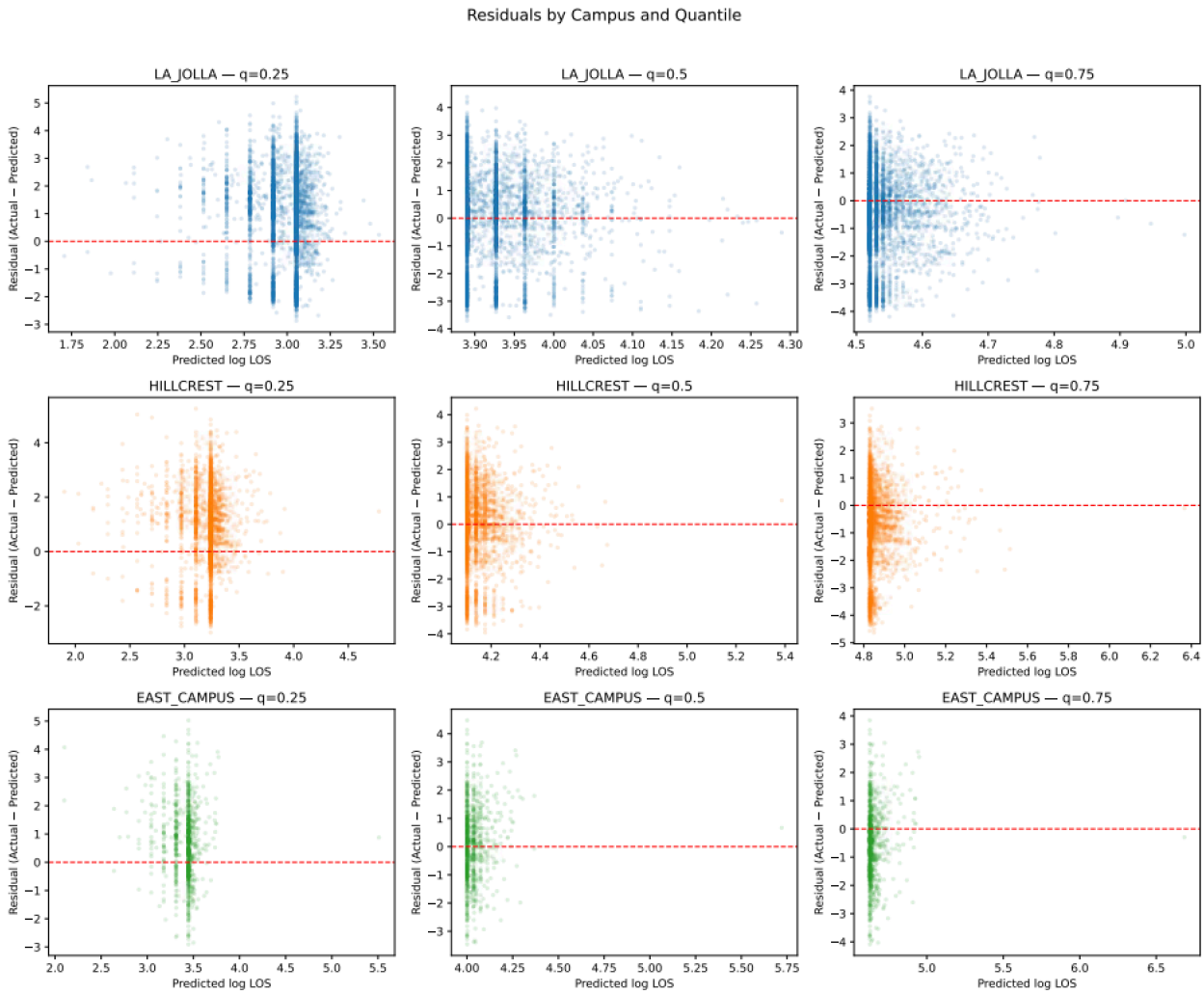


Figure 1: Residual plots (actual minus predicted log LOS) by campus and quantile. Each column corresponds to a quantile ($q \in \{0.25, 0.50, 0.75\}$) and each row to a campus (La Jolla, Hillcrest, East Campus).

## 3.3 Baseline Simulation Results

Figures 2–4 show smoothed daily average bed utilization over the six-month constrained baseline simulation for each campus. All three campuses exhibit a warm-up period during

the first 25–30 days as the system populates from an empty state, after which utilization stabilizes into steady-state fluctuations. Floor beds dominate utilization at all three campuses, with La Jolla stabilizing around 340 beds, Hillcrest around 130–150 beds, and East Campus around 82 beds. ICU and boarding resources stabilize at lower absolute levels but represent high-acuity and operationally critical constraints. ED utilization remains low and stable across all campuses, confirming that emergency department capacity is not the primary bottleneck in the system.

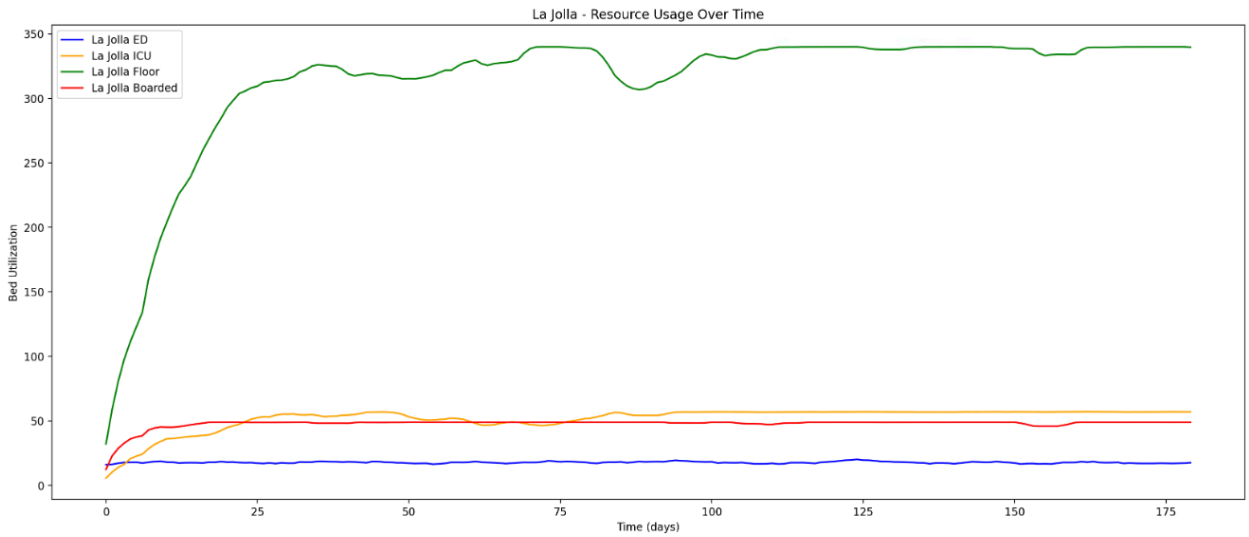


Figure 2: La Jolla average daily resource utilization over the six-month baseline simulation.

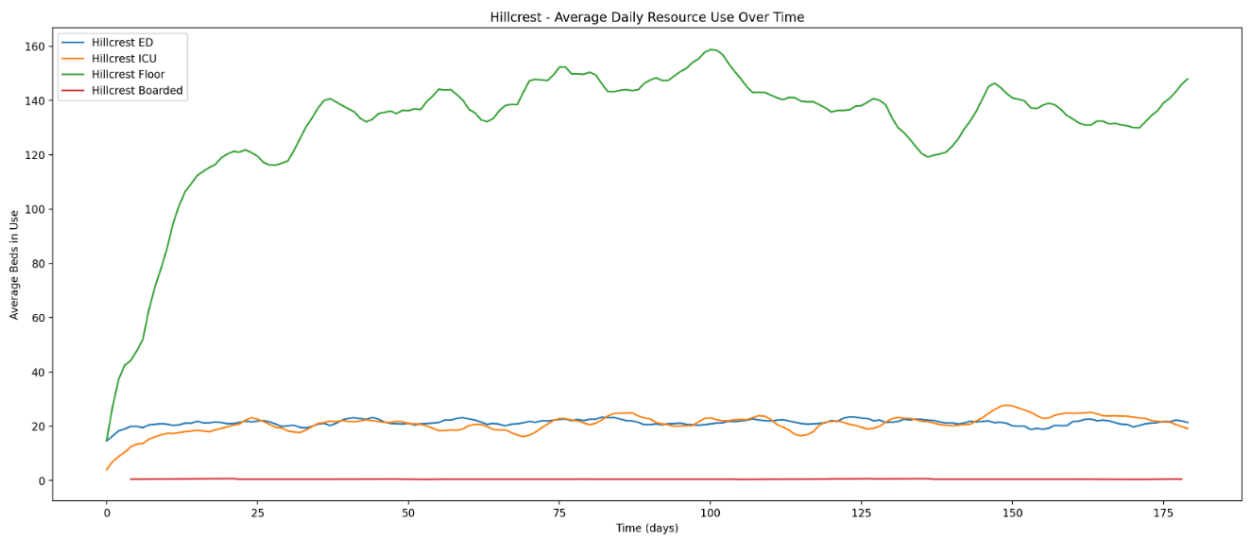


Figure 3: Hillcrest average daily resource utilization over the six-month baseline simulation.

Figure 5 shows the proportion of simulation time each resource spent above 80% utilization. La Jolla resources are the most chronically constrained: the floor unit exceeded 80% utilization 73% of the time, the ICU 64% of the time, and the boarding unit 59% of the time. East Campus shows similarly high constraint across ICU (63%), floor (64%), and

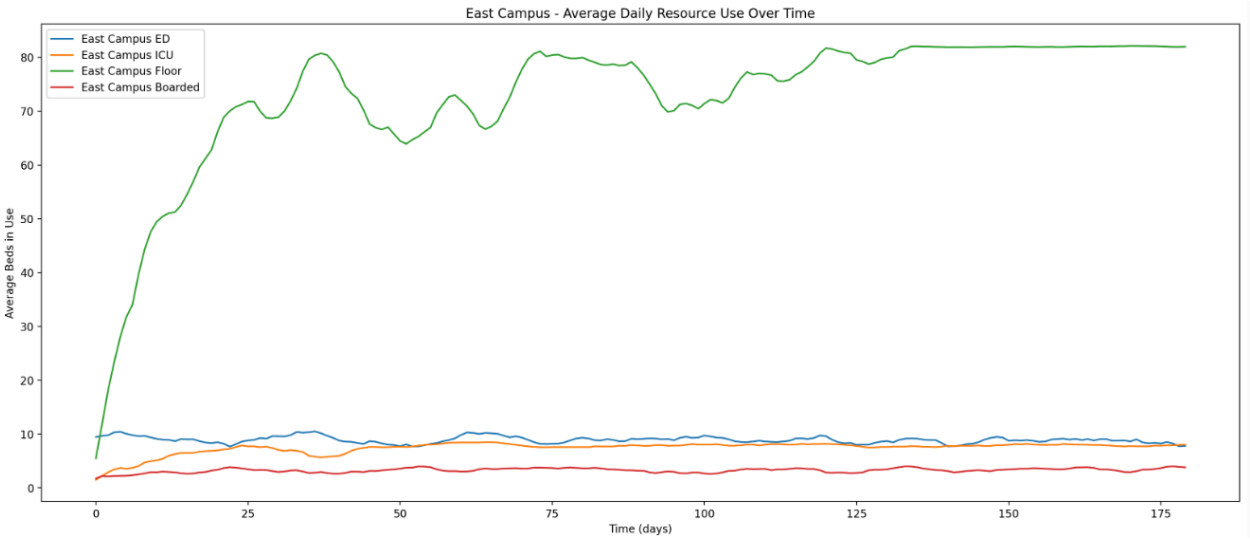


Figure 4: East Campus average daily resource utilization over the six-month baseline simulation.

boarding (54%) resources, despite being the smallest campus. Hillcrest resources spend comparatively less time above 80% utilization, suggesting greater available slack capacity. ED resources across all campuses spend near-zero time above 80% utilization, reinforcing that inpatient capacity—not emergency department throughput—is the binding constraint in this system.

Figure 6 shows average wait time per resource among all queue events. ICU_EAST_CAMPUS produces by far the longest average wait at approximately 220 hours, reflecting the catastrophic queuing dynamics that emerge when a resource with only eight beds reaches capacity. ICU_LA_JOLLA shows the second-longest average wait at approximately 45 hours. Boarding and floor resources at La Jolla show moderate average waits of 15–17 hours. ED resources and Hillcrest inpatient resources show near-zero average waits, consistent with their lower utilization rates. Frequency of delay follows a similar pattern: BOARDED_LA_JOLLA experienced a non-zero wait in approximately 89% of queue events, and ICU_EAST_CAMPUS and ICU_LA_JOLLA also showed high wait rates, while Hillcrest and ED resources across all campuses were rarely delayed.

These findings motivate the transfer policy experiments described in Section 3.4, which evaluate whether LOS-based early transfers can meaningfully reduce congestion at these resources.

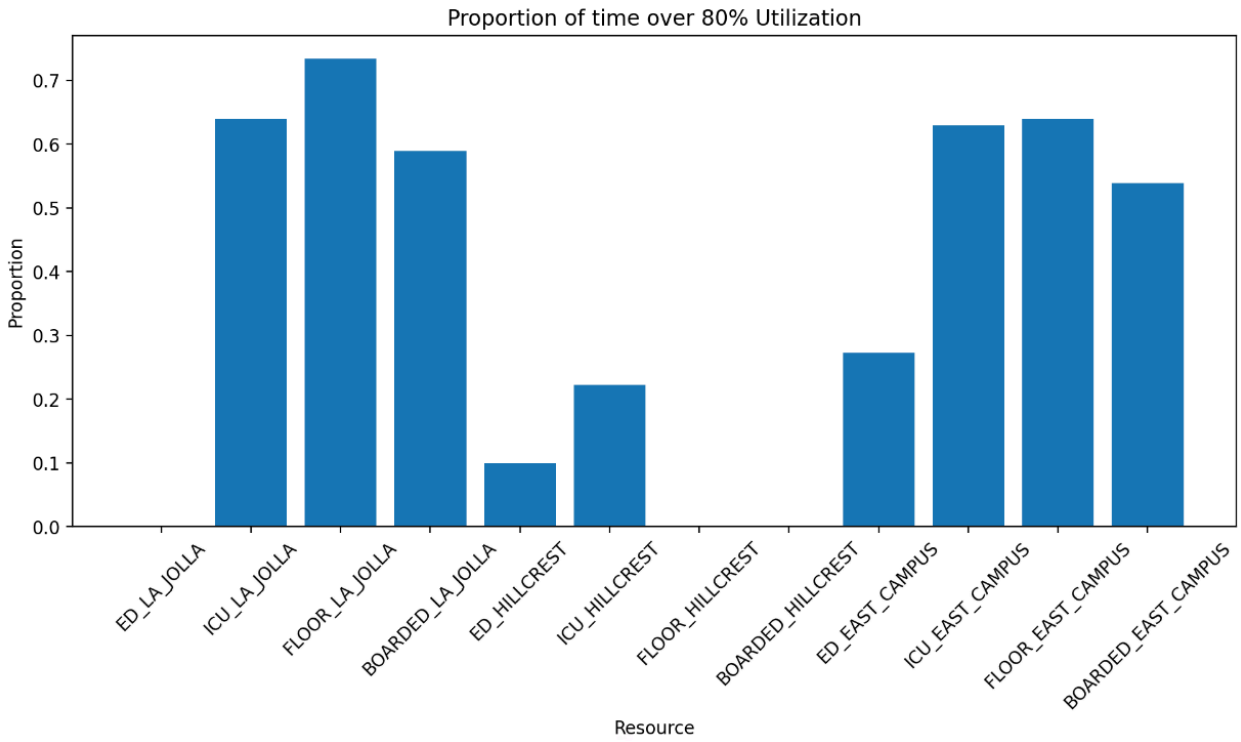


Figure 5: Proportion of simulation time each resource spent above 80% utilization in the baseline constrained run.

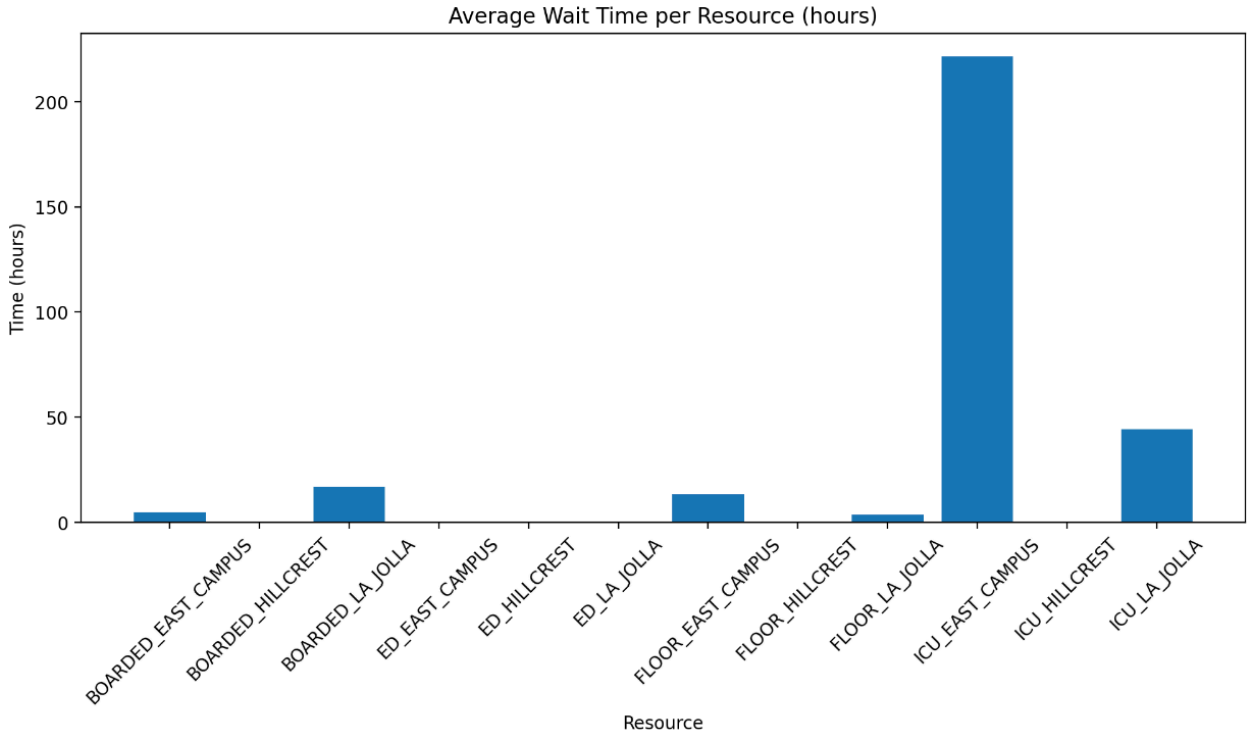


Figure 6: Average wait time per resource (hours) among all queue events in the baseline simulation.

## 3.4 Transfer Policy Results

Three separate transfer policy experiments were conducted, each designating one campus as the source of LOS-triggered transfers. In each experiment, patients admitted to the source campus whose predicted LOS at the 75th quantile exceeded a threshold of 3, 8, or 12 hours were rerouted at the moment of ED admission to one of the two receiving campuses. Results are compared against the no-transfer baseline described in Section 3.3.

### 3.4.1 Effect on Source Campus Utilization

Figures 7–9 show smoothed daily bed utilization at the boarding unit and ICU of each source campus under the baseline and three transfer thresholds.

Transferring patients out of La Jolla produced the most dramatic reduction in congestion. Under all three thresholds, La Jolla's boarding unit collapsed from a baseline peak of 49 beds to near zero within the first several days of the simulation and remained there for the full six-month run. La Jolla's ICU, which peaked at 57 beds under baseline conditions, declined to the 40–46 bed range under transfer policy runs, with the 12-hour threshold producing the most sustained reduction below the 80% capacity line of 46 beds.

Transferring patients out of Hillcrest produced similarly complete relief of its boarding unit, which dropped from a baseline of approximately 10 beds to near zero across all thresholds. Hillcrest's ICU showed more variable but meaningful improvement: the baseline saturated at the stated capacity of 30 beds for the majority of the simulation, while all three threshold runs reduced mean daily ICU occupancy to the 20–28 bed range, consistently below the 80% capacity threshold of 24 beds.

Transferring patients out of East Campus reduced its boarding unit from a baseline peak of 4 beds to near zero under all thresholds. East Campus ICU, which reached a baseline peak of 8 beds at full capacity, declined substantially under all three threshold conditions, with the 12-hour threshold producing the clearest sustained separation below the 80% capacity line.

### 3.4.2 Wait Time Distributions

Figure 10 shows the distribution of wait times at the most constrained resource for each source campus experiment. For La Jolla's boarding unit, the baseline distribution is right-skewed with a peak near 5–10 hours and a long tail extending beyond 25 hours. Under all three transfer thresholds, near-zero wait events were recorded at this resource, as patients predicted to require long stays were rerouted before reaching the boarding unit. For Hillcrest's ICU and East Campus's ICU, the baseline distributions are broad and heavy-tailed, consistent with the catastrophic queuing dynamics observed in Section 3.3. Under the 3-hour threshold, both distributions shift sharply left and compress, with nearly all wait events concentrated below 25 hours. The 8-hour and 12-hour thresholds produce progressively wider distributions, reflecting fewer transfers and correspondingly less relief, though both

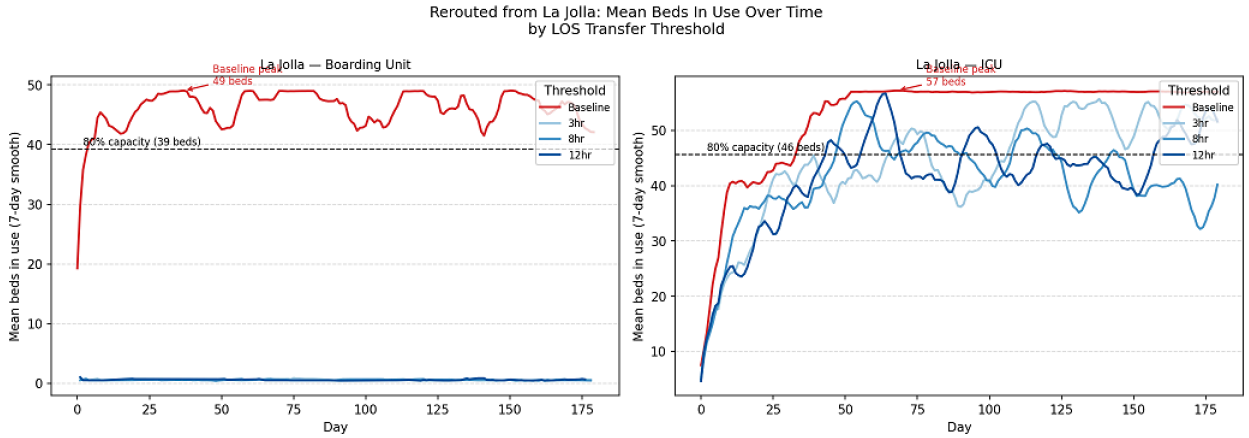


Figure 7: La Jolla boarding unit and ICU mean daily bed utilization under the no-transfer baseline and three LOS transfer thresholds. Dashed line indicates 80% of stated capacity. All three thresholds eliminate boarding congestion and meaningfully reduce ICU occupancy relative to baseline.

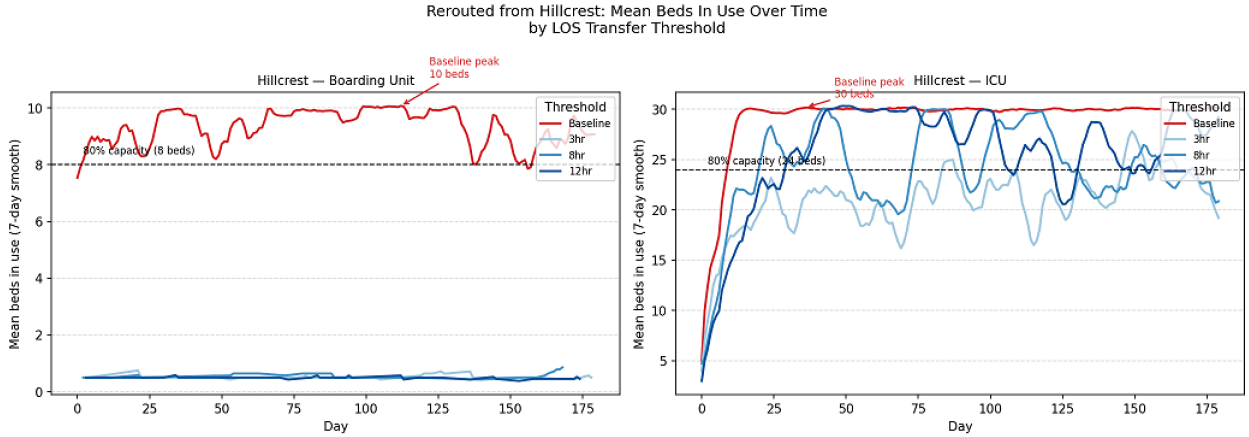


Figure 8: Hillcrest boarding unit and ICU mean daily bed utilization under the no-transfer baseline and three LOS transfer thresholds. Boarding congestion is eliminated under all thresholds. ICU occupancy is reduced from capacity-saturated baseline levels, with variability increasing at more conservative thresholds.

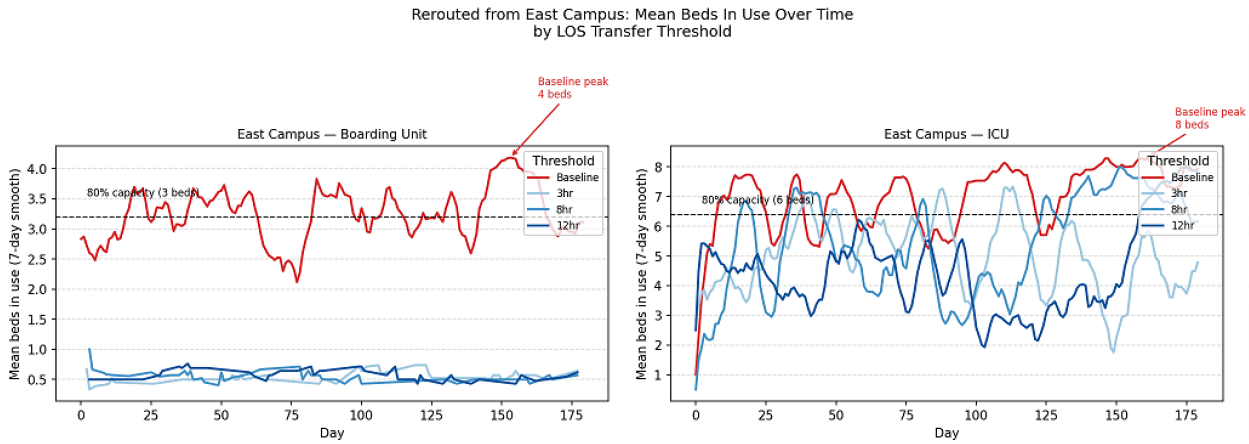


Figure 9: East Campus boarding unit and ICU mean daily bed utilization under the no-transfer baseline and three LOS transfer thresholds. Despite East Campus having the smallest absolute capacity, all three thresholds substantially reduce both boarding and ICU utilization relative to baseline.

remain substantially improved relative to baseline.

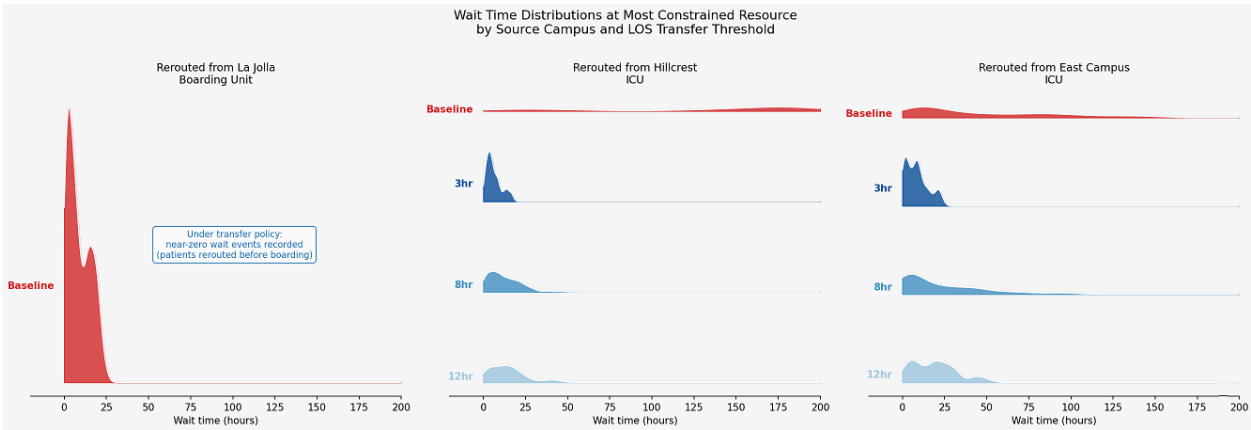


Figure 10: Wait time distributions at the most constrained resource for each source campus experiment. Left panel: La Jolla boarding unit — transfer policy runs produce near-zero wait events. Center and right panels: Hillcrest ICU and East Campus ICU — baseline distributions are broad and heavy-tailed; more aggressive thresholds shift distributions leftward and compress the tail.

### 3.4.3 Sensitivity to Threshold and Load-Shifting Effects

Figure 11 shows how mean wait time at the two most constrained system resources responds to transfer threshold across all three source campus experiments. Two distinct patterns emerge.

At La Jolla's boarding unit (left panel), transferring patients out of La Jolla (blue line) reduces mean boarding wait to near zero at the 3-hour threshold and maintains it there at

8 and 12 hours. However, transferring patients out of Hillcrest or East Campus increases mean wait time at La Jolla's boarding unit relative to baseline, rising to approximately 17–20 hours at the 3-hour threshold before partially recovering at more conservative thresholds. This load-shifting effect indicates that patients rerouted away from Hillcrest and East Campus are being received at La Jolla, stressing its boarding capacity.

At East Campus's ICU (right panel), transferring patients out of East Campus (red line) reduces mean ICU wait to near zero across all thresholds. Transferring patients out of La Jolla produces moderate increases in East Campus ICU wait, remaining below 50 hours. The most striking pattern is the Hillcrest line (green): at the 3-hour threshold, mean wait time at East Campus ICU spikes to approximately 220 hours before declining to 80 hours at 8 hours and rising again to 165 hours at 12 hours. This non-monotonic response suggests that the most aggressive Hillcrest threshold overwhelms East Campus ICU capacity before the system partially stabilizes at intermediate thresholds.

These findings demonstrate that LOS-triggered transfers reliably reduce congestion at the source campus under all threshold conditions tested. However, the system-level benefit depends critically on which campus serves as the source: La Jolla transfers produce unambiguous system-wide improvement, while transfers originating from smaller campuses can introduce or amplify bottlenecks at receiving facilities, particularly at East Campus's eight-bed ICU.

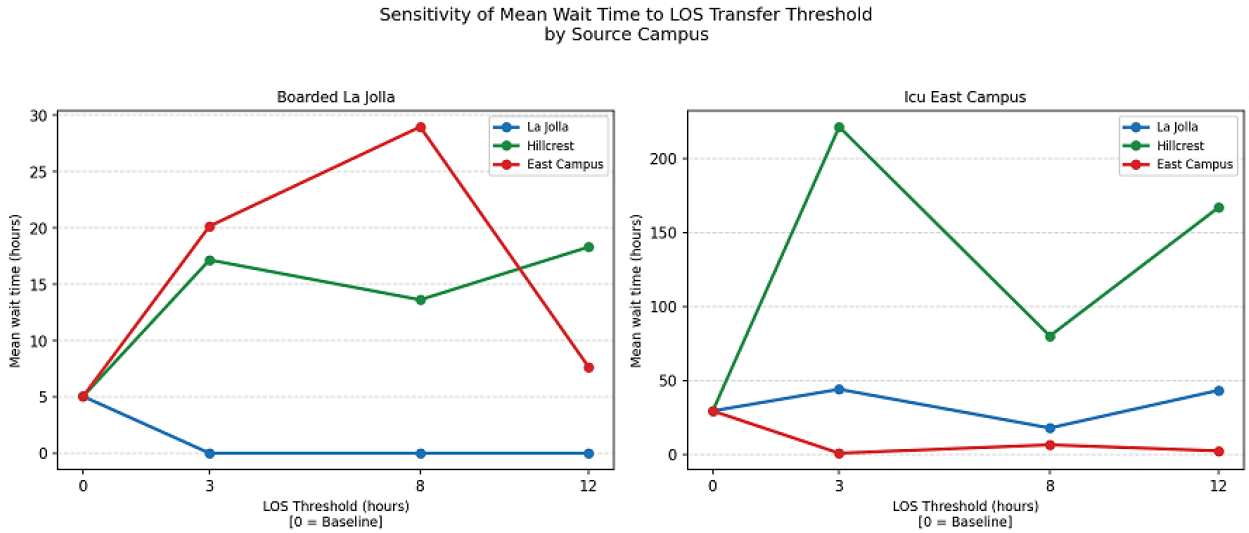


Figure 11: Sensitivity of mean wait time to LOS transfer threshold at La Jolla's boarding unit (left) and East Campus ICU (right), by source campus. Each line represents a separate transfer experiment. Transferring from La Jolla eliminates boarding waits; transferring from other campuses loads La Jolla as a receiving facility. The Hillcrest-to-East-Campus pathway produces a non-monotonic spike at the 3-hour threshold, suggesting East Campus ICU is overwhelmed under aggressive rerouting from Hillcrest.

# 4 Discussion

This project demonstrates that a trajectory-driven discrete event simulation calibrated to institutional ADT data can faithfully reproduce operational patterns in a multi-campus health system and serve as a testbed for evaluating transfer policies that would be impractical to pilot in a live clinical environment. Three findings from the transfer policy experiments warrant discussion.

**LOS-triggered transfers are effective at the source campus but do not uniformly improve the system.** Under all three threshold conditions and across all three source campuses, LOS-triggered transfers at the moment of ED admission reliably eliminated boarding congestion and substantially reduced ICU occupancy at the source facility. This is consistent with prior DES literature demonstrating that early intervention at the point of admission, rather than reactive bed management after congestion has developed, produces the most durable reductions in wait times. Ahmed and Alkhamis (2009) demonstrated that strategic resource allocation at the point of patient entry could achieve substantial reductions in wait time without increasing costs. However, the system-level effect was asymmetric. Transfers originating from La Jolla, the highest-volume campus, produced improvements at La Jolla without materially degrading conditions at receiving campuses. Transfers originating from Hillcrest and East Campus, by contrast, introduced load-shifting effects that elevated wait times at La Jolla's boarding unit and, in the case of aggressive Hillcrest transfers, catastrophically overloaded East Campus's eight-bed ICU. This asymmetry reflects a fundamental capacity imbalance in the system: La Jolla generates sufficient patient volume that redistributing its long-stay patients produces meaningful relief at a scale the receiving campuses can absorb, while smaller campuses lack the volume to drive comparable improvements and their transfers instead stress already-constrained receiving resources.

**Threshold aggressiveness interacts nonlinearly with receiving campus capacity.** The sensitivity analysis revealed a non-monotonic response at East Campus ICU when transfers originated from Hillcrest: mean wait time spiked to approximately 220 hours at the 3-hour threshold before declining at 8 hours and rising again at 12 hours. This behavior is consistent with the well-documented instability of small-capacity queuing systems under sudden demand increases. East Campus ICU has only eight beds, placing it in a regime where even modest additional patient load can trigger catastrophic queue buildup. The implication for operational policy is that threshold selection cannot be made on the basis of source campus outcomes alone: a threshold that appears effective when evaluated at the source may simultaneously overwhelm a receiving facility. Any deployment of LOS-triggered transfer logic in a real health system would require real-time visibility into receiving campus capacity before a transfer is authorized.

**Predictor limitations constrain the precision of the transfer trigger.** The quantile regression LOS predictor uses only three features available at the moment of ED admission:

time in ED, number of prior admissions in the preceding 90 days, and admitting campus. The residual plots in Section 3.2 show wide spread across all campuses and quantiles, reflecting the inherent difficulty of predicting LOS from admission-time information alone. The vertical banding pattern in predicted values indicates limited resolution, with the model producing a small number of distinct predicted LOS values rather than a continuous distribution. A more accurate predictor would reduce the number of patients that need to be transferred to achieve a given reduction in congestion. Future work integrating richer clinical features — acuity scores, presenting diagnosis, or real-time lab values — could substantially improve predictor resolution and allow more targeted transfer triggering with lower operational burden on receiving campuses.

**Limitations.** Several limitations of this study should be acknowledged. First, the simulation models each source campus experiment in isolation: in each run, only one campus's patients are subject to transfer logic, while the other two campuses operate as in the baseline. Real-world deployment would involve simultaneous transfer decisions across all campuses, producing interaction effects not captured here. Second, the transfer rerouting logic generates new patient trajectories stochastically from the empirical transition matrix and fitted LOS distributions rather than assigning real historical trajectories, introducing model uncertainty in the post-transfer portion of each patient's path. Third, the simulation does not model staffing constraints, patient consent, or clinical contraindications to transfer, all of which would reduce the feasible transfer rate in practice. Finally, capacity parameters were derived from steady-state occupancy in an unconstrained simulation rather than from licensed bed counts, which were unavailable at the tier level; while this approach produces operationally consistent parameters, it may not reflect licensed capacity constraints during surge conditions.

# 5 Conclusion

This project developed a multi-facility discrete event simulation of patient flow across UC San Diego Health's three campuses, driven by a trajectory library of 150,337 real patient journeys extracted from institutional ADT data. The simulation reproduces operational utilization patterns consistent with known system characteristics and identifies La Jolla and East Campus as the most chronically constrained facilities under baseline conditions. Integration of a quantile regression LOS predictor as a transfer trigger demonstrates that LOS-based early transfers can substantially reduce boarding congestion and ICU pressure at the source campus across a range of threshold conditions. However, the system-level viability of any transfer policy depends on the capacity of receiving facilities to absorb redirected patients, and aggressive thresholding from smaller campuses can introduce or amplify bottlenecks at receiving sites.

These findings suggest that a La Jolla-first transfer policy, in which long-stay patients predicted at ED admission are selectively rerouted to Hillcrest or East Campus, represents the most promising near-term operational intervention for UCSD Health. The 8- and 12-hour

thresholds appear to offer the best balance between source campus relief and receiving campus stability, avoiding the queue instability observed at East Campus ICU under the 3-hour threshold.

Future work should pursue three directions. First, expanding the predictor feature set with clinical variables available at the time of admission could improve LOS prediction resolution and reduce the transfer volume required to achieve meaningful congestion relief. Second, modeling simultaneous multi-campus transfer logic would capture interaction effects between concurrent transfer decisions and provide a more realistic assessment of system-wide policy impact. Third, incorporating real-time capacity visibility into the transfer trigger — so that transfers are only authorized when the receiving campus has available capacity — would address the load-shifting instability observed in the sensitivity analysis and bring the simulation logic closer to clinically deployable decision support.

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